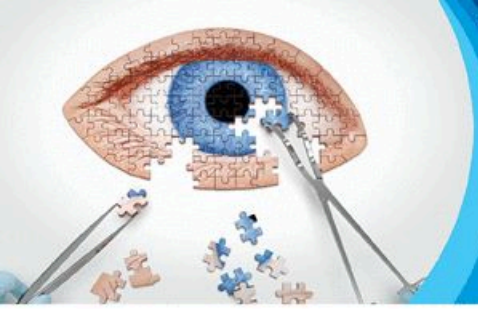


EXT
YOUR
A guide to modern cata

"Citius, Altius, Fortius" - Faster, higher, stronger – this well-recognized Olympian motto also encapsulates the constant human endeavor towards improvement and progress, trying to make the future better than the present. It encompasses all spheres of life, such as technology, especially in healthcare.

In the field of ophthalmology, the technology involved in cataract surgery procedures and the lens implant has advanced by leaps and bounds, pole-vaulting consistently through greater heights. From mere vision restoration to vision enhancement, from key-hole size incisions to pin-hole size incisions, from weeks of rest to just a daycare procedure, cataract surgery has traversed a significant distance.

In this read, we will take a tour of a very important aspect of cataract surgery: The choice of lens implant.

What is a cataract?

Cataract is the opacification of the natural crystalline lens in the eye primarily due to aging. As cataract envelops the lens, it clouds the vision. Cataract patients find that their vision is blurry, dull colours and also deteriorating night vision.

What is a lens implant?

During cataract surgery, the natural lens affected by the cataract is removed and is replaced by an artificial lens implant, also referred to as IOL (Intra Ocular Lens).

What are the different types of lens implant options?

- **Monofocal lens :** Designed to address distance vision.
- **Toric lens :** Designed to address distance vision and pre-existing astigmatism.
- **Presbyopia correcting lens :** Designed to address near, intermediate and distance (Primarily of two types, Trifocal and Extended Vision lens).

How does a Monofocal lens work?

As the name implies, the Monofocal lens focuses light rays at a single point and supports distance vision only. For vision at near and intermediate distances, one would need glasses.

Monofocal lens implants are of two design types. Spherical and Aspheric design. The Aspheric lens design has an edge over spherical design from an image sharpness stand point especially in low light condition.

Patient Profile: An MSME business owner in his mid-fifties, with two units in nearby cities, needs to constantly shuttle between his units. He mostly prefers driving at night to avoid traffic and save time. What type of lens implant would be most suitable for this cataract patient?

Aspheric Monofocal would be optimal for this patient profile as it would enable him to have improved distance vision without glasses. It would support his night driving as there would be less visual disturbances such as glare or haloes; however, he will continue to depend on glasses for near and intermediate activities.

MODERN CATARACT LENS OPTIONS

Lens type	Near	Mid-range	Distance	Astigmatism correcting
Monofocal			Clear vision	—
Astigmatism Correcting lens (Toric)			Clear vision	Clear vision
Presbyopia Correcting lens (Extended Vision lens / Trifocal lens)	Clear vision	Clear vision	Clear vision	—
Presbyopia Correcting Toric lens (Extended Vision lens / Trifocal lens)	Clear vision	Clear vision	Clear vision	Clear vision

What is Astigmatism and how is it addressed?

Cataract patients may also have pre-existing astigmatism, a visual disorder where the cornea is irregularly shaped in the form of a rugby ball, causing distorted vision. Such patients can opt for astigmatism correcting lens (TORIC) or presbyopia plus astigmatism correcting lens implant.

Patient Profile: A journalist with pre-existing astigmatism is about to undergo cataract surgery. Is there a lens implant that can help her condition without the need for spectacles?